Raqib: every HR transaction, checked against every rule

Continuous policy assurance for HR. Each exception explained with its rule and evidence, routed to an owner, and decided by a person.

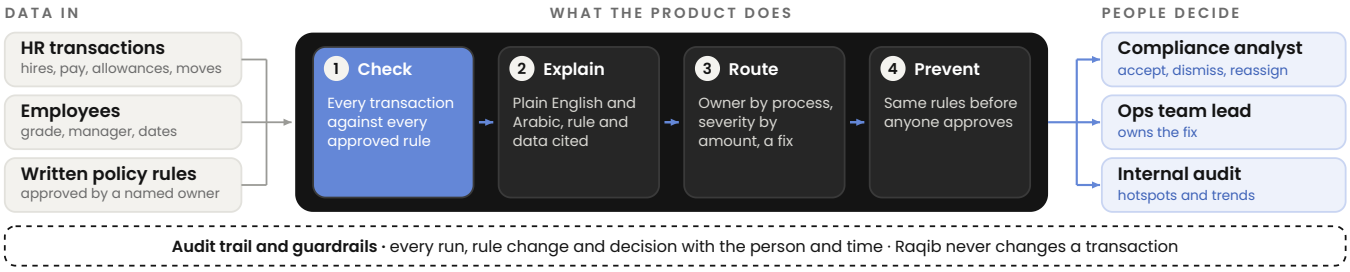
440/440	0	0/170	52 ms
planted violations found	false positives	near misses wrongly flagged	9,744 transactions × 18 rules

Data limitation. The official data pack and test file were not published, so we curated a synthetic sample in the brief's shape (2,000 employees, 9,744 transactions, seed 2310, never used while building). Results show the approach on that sample, not on Emirates data. **All findings and files:** github.com/pratikkp24/aonami-raqib-emirates-hr-compliance

Try the live prototype
aonami-raqib.vercel.app
Hosted, no install. Press **Guided demo**.
Demo video loom.com/share/...

THE APPROACH

Compliance today relies on sampling, so issues surface late and coverage is uneven. Raqib replaces sampling with a **deterministic rules engine** that checks every transaction against every approved rule, then explains, routes and records each exception for a human decision. **No AI model decides whether something is a breach**: the same batch always gives the same findings, and each one can be traced to a rule and the exact data that broke it.



DETECTION ACCURACY

Before the official pack, we built our own: 2,000 employees and about 9,800 transactions over 12 months, 18 rules across the four named families, about 440 planted violations, plus **170 deliberate near misses** (exactly at a limit, one day inside a window, approved on the effective date) that must not be flagged.

BENCHMARK	RESULT
Precision, three generated packs	99.8% to 100%
Recall, three generated packs	100%
Near misses wrongly flagged	0 of 170

Unusual patterns that no rule covers (outlier amounts, out-of-hours approvals) are kept in a **separate, unscored lane**, so they never add false positives.

EXPLANATION AND ROUTING

Every finding carries the rule, a one-line reason, the evidence, an owner, a severity and a corrective action:

```
R15 · Split to avoid approval limit · High · Payroll
Operations
With TX009158 within 30 days, the combined AED 9,100
needs grade 11; approved at grade 8.
Evidence: related=TX009158 (AED 5,000); combined=AED
9,100; required grade=11+; approver grade=8
Fix: treat the changes as one and obtain approval at
the combined amount.
```

Owners follow the process: Payroll Operations, Talent Acquisition, HR Mobility, Time and Attendance, Offboarding. **Severity** starts from the rule and rises with materiality (for example, the top approval tier). The analyst works a queue sorted by severity and amount; team leads see their own queue; the audit lead sees hotspots by process, team and rule, and monthly trends.

ADDING RULES WITHOUT CODE

A policy owner types a rule in plain English, for example *"Allowances above AED 15,000 need an approver of grade 13 or above"*. Raqib matches it to a known check type and shows the exact parameters. It runs only after a named person approves it, and the change is recorded. This is pattern matching, not a language model, so it is predictable and reviewable.

FIT FOR EMIRATES AND OTHER GOVERNMENT ENTITIES

Rules are data, not code, so another entity loads its own policy and approval matrix without changing the engine. The interface is fully bilingual (English and Arabic, right to left). The same engine runs in the browser and from the command line for scheduled monthly batches.

HOW IT RUNS INSIDE EMIRATES' ENVIRONMENT

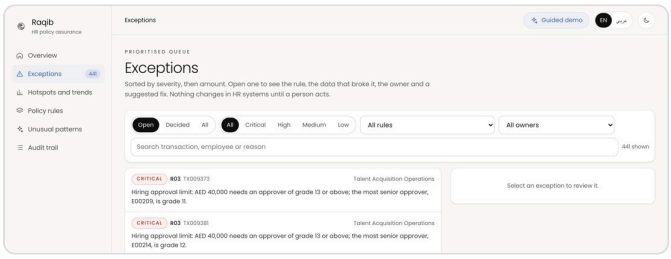
- **Deployment:** a static web application and a command-line runner, packaged as a container and deployed in Emirates' own environment (private cloud or on premises). No outbound connection is needed.
- **Data in:** read-only exports or API pulls from the HR system of record (transactions, employees, approval matrix). Raqib never writes back to HR systems.
- **Access:** Emirates single sign-on with roles: compliance analyst, operations team lead, internal audit (read only). Decisions carry the signed-in person's name.
- **Outputs:** findings in the agreed audit finding format, exported to the case or ticketing tool Emirates already uses, plus the audit trail.

RESPONSIBLE AI AND SECURITY STATEMENT

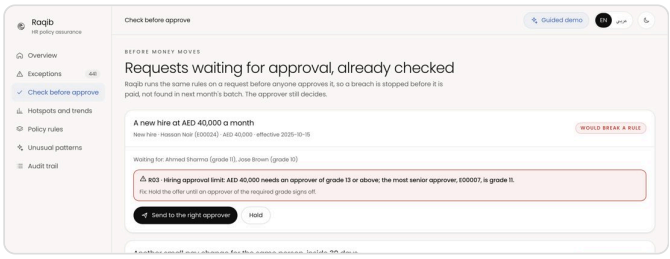
Raqib recommends; a person decides. It never changes a transaction. Every exception needs a recorded human decision (accept, dismiss with a reason, or reassign), and every run, rule change and decision is logged with the person and the time.

No AI model touches the data. Detection is deterministic and explainable, every flag is traceable to a rule and the data behind it, and the synthetic pack was processed only on our own machine: it was not uploaded to any service that keeps or trains on data. The hosted demo runs on Aonami's own synthetic sample, and files loaded there stay in the viewer's browser.

- **Fairness and employee experience:** wording is neutral ("does not match rule R07"), not accusatory. Hotspots are shown by process and team, never by individual. In a pilot, employees affected by a correction are told what changed and why, and can respond before any recovery action.
- **Data protection:** designed for the UAE Personal Data Protection Law (Federal Decree-Law 45 of 2021): minimal fields, data stays in Emirates' environment, and audit records hold IDs rather than personal details. It follows the principles of the UAE Charter for the Development and Use of AI: transparency, human oversight and accountability.
- **Limits we disclose:** results above are on our synthetic benchmark; accuracy on Emirates' rules depends on agreeing the exact meaning of each written rule with policy owners, which is the first step of a pilot. Arabic text needs native review before production.



Every exception explained. The rule, the exact data that broke it, the owner and the fix; a person accepts, dismisses with a reason or reassigns.



Check before approve. The same rules run before anyone signs off, so a breach is stopped before it is paid.

PILOT NEXT QUARTER

WEEKS	ACTIVITY
1 to 2	Load the policy and approval matrix; agree each rule's meaning with policy owners
3 to 4	Shadow run on last quarter's transactions; compare with the sampling results
5 to 6	Live monthly batch with the compliance team; tune severity and routing
End	Coverage, exceptions found, time to decide, and a go or no-go

TEAM

Aonami builds AI agents and decision systems for regulated institutions, with a named person approving every action. A team of nine across engineering, AI and delivery; co-founders Pratik Kumar and Jayesh Kumar Rout, with backgrounds in lending, fintech and risk at Stashfin, OneCard and Northern Trust.

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